Lab Thirteen – Statistical Inference

1 Remaining Portfolio Work!

By 5p 4/27 Email me a link to your GitHub page with all polished labs rendered, and your review material (at least one unit). In the email, you should include a reflection about the processes involved in creating this portfolio and what you learned as a result. If you're having trouble reflecting, consider Fink's Taxonomy of Significant Learning (Google Fink taxonomy aligned self-reflection questions). In the email, let me know whether you would be comfortable sharing your page – I'm hoping that together we can get a good study guide for the final.

2 Statistical Inference

Kasdin et al. (2025) show that dopamine in the brains of young zebra finches acts as a learning signal, increasing when they sing closer to their adult song and decreasing when they sing further away, effectively guiding their vocal development through trial-and-error. This suggests that complex natural behaviors, like learning to sing, are shaped by dopamine-driven reinforcement learning, similar to how artificial intelligence learns. You can find the paper at this link: https://www.nature.com/articles/s41586-025-08729-1.

Note that the measurement is rather complicated. First, the measurements are taken on syllables of a bird song. When discussing bird songs, the term syllable refers to a single, distinct, uninterrupted sound that is part of a possibly complex bird song.

They initially describe their measurement as follows. They compare the sound of the developing zebra finches to the median sound of adult zebra finches, that act as a "goal" sound.

> "For each syllable rendition, we calculated the proximity (Euclidean distance) to the median of the adult renditions of the syllable (age > 90 days) in feature space (Fig. 2a,b)."

They contextualize that measurement by computing a "relative quality" that takes into account their current ability (e.g., are they getting closer or further compared to their previous 14 renditions).

> "Because dopamine is hypothesized to encode the relative value of performance, we defined the 'relative quality' of a syllable as proximity to the adult version of the syllable relative to the mean of the previous 14 renditions (Methods)."

They define the "closer" measurements as the 10% of syllable renditions with the highest relative quality and the "further" as the lowest 10%.

> "To test for online error responses, for each day we compared the dopamine activity associated with the
> 10

Note that the data are measures of dopamine change, they use fibre photometry, changes in the fluorescence indicate dopamine changes in realtime. Their specific measurement considers the percent change in flourescence in 100-ms windows between 200 and 300 ms from the start of singing, averaged across development.

Finally, the researchers note that they adjust the observations to correct for the increased dopamine for singing at all (whether it is close or far).

> "As previously reported, the baseline level of dopamine in Area X is known to increase during singing; to isolate the effect of syllable quality from these singing-related dopamine increases, we subtracted the mean dopamine response across all renditions of a syllable on each day from the dopamine response for each rendition."

So, the random variables we are working with are:

$$\begin{aligned}
X_{\text{close}} &= \text{average \% change in fluorescence for the top 10\% syllable renditions} \\
&\quad \text{relative to distance from adult sound (corrected for singing)} \\
X_{\text{far}} &= \text{average \% change in fluorescence for the bottom 10\% syllable renditions} \\
&\quad \text{relative to distance from adult sound (corrected for singing)}
\end{aligned}$$

That is, the unit of measure is the syllable as measured on six birds, and the measurement is somewhat complicated.

2.1 Part a

Using the `pwr` package for `R` (Champely 2020), conduct a power analysis. How many observations would the researchers need to detect a moderate-to-large effect ($d = 0.65$) when using $\alpha = 0.05$ and default power (0.80) for a two-sided one sample t test.

Solution:

Installed and loaded the `pwr` library and used the `pwr.t.test` to find that the number of observations needed (n) would be 21.

```r
library(pwr) # load package

pwr.t.test(
  d = 0.65,
  power = 0.80,
  alternative = "two.sided",
  type = "one.sample"
)
```

```
     One-sample t test power calculation

              n = 20.58039
              d = 0.65
      sig.level = 0.05
          power = 0.8
    alternative = two.sided
```

2.2 Part b

Click the link to go to the paper. Find the source data for Figure 2. Download the Excel file. Describe what you needed to do to collect the data for Figure 2(g). Note that you only need the `closer_vals` and `further_vals`. Ensure to `mutate()` the data to get a difference (e.g., `closer_vals - further_vals`).

Solution:

1. Downloaded the figure 2 excel source data
2. Copy-pasted the `Closer_vals` and `Further_vals` to a new excel sheet and created column labels for them
3. Saved the excel sheet as a csv file in my project's root folder
4. Used `mutate()` to get a new column for difference

```r
fig.dat <- read_csv("fig2_dat.csv")

fig.dat <- fig.dat |>
  mutate(Difference = Closer - Further)

head(fig.dat)
```

```
# A tibble: 6 x 3
  Closer Further Difference
   <dbl>   <dbl>      <dbl>
1 0.356  -0.224       0.580
2 0.0542 -0.107       0.161
3 0.144  -0.187       0.331
4 0.122  -0.110       0.233
5 0.109  -0.170       0.279
6 0.318  -0.0897      0.408
```

2.3 Part c

Summarize the data.

- Summarize the further data. Do the data suggest that dopamine in the brains of young zebra finches decreases when they sing further away?
- Summarize the closer data. Do the data suggest that dopamine in the brains of young zebra finches increases when they sing closer to their adult song?
- Summarize the paired differences. Do the data suggest that there is a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song?
- **Optional Challenge** Can you reproduce Figure 2(g)? Note that the you can use `geom_errorbar()` to plot the range created by adding the mean $\pm$ one standard deviation.

Solution:

Created numerical and graphical summaries for further, closer, and paired differences data.

- The mean for the further condition is negative (-0.213), suggesting that dopamine levels tend to decrease when the birds sing further away from their adult song

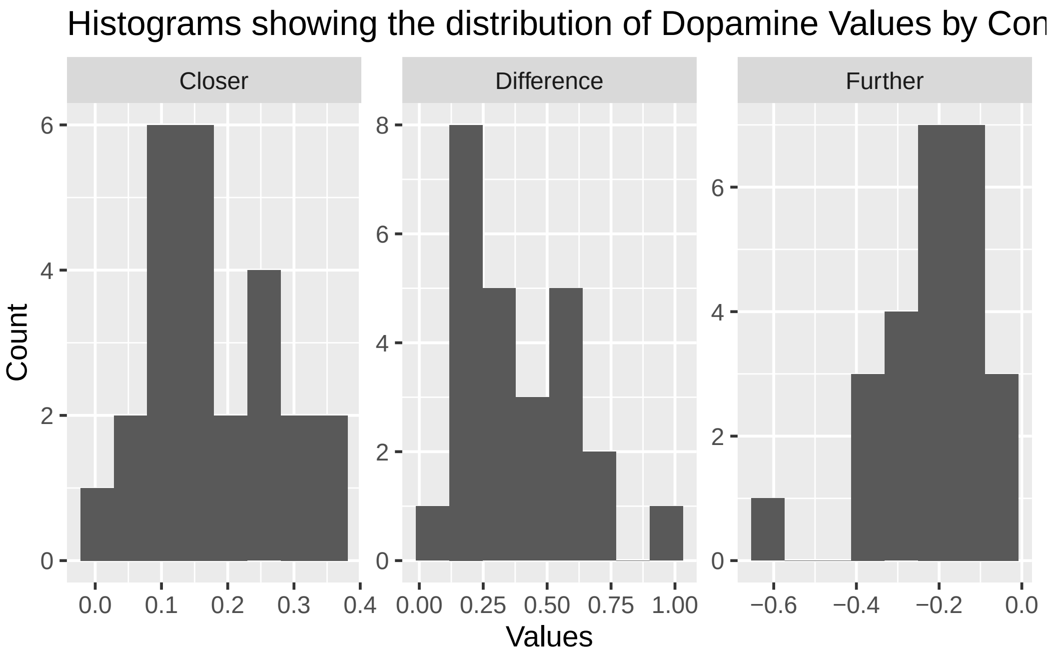
- The mean for the closer condition is positive (0.179), indicating that dopamine levels tend to increase when the birds sing closer to their adult song

- The mean for difference is positive (0.391) and larger than for closer, showing that dopamine levels are generally higher when birds sing closer compared to further away

```r
# reshape with pivot_longer
fig.long <- fig.dat |>
  pivot_longer(
    cols = everything(),
    names_to = "Condition",
    values_to = "Value"
  ) |>
  drop_na()

# numerical summaries
fig.summary <- fig.long |>
  group_by(Condition) |>
  summarize(
    Mean = mean(Value),
    SD = sd(Value),
    Median = median(Value),
    IQR = IQR(Value)
  )

fig.summary
```

```
# A tibble: 3 x 5
  Condition    Mean     SD Median   IQR
  <chr>       <dbl>  <dbl>  <dbl> <dbl>
1 Closer      0.179 0.0975  0.144 0.142
2 Difference  0.391 0.214   0.353 0.305
3 Further    -0.213 0.134  -0.188 0.191
```

```r
# graphical summary
fig.plot <- ggplot(fig.long) +
  geom_histogram(aes(x = Value), bins = 8) +
  facet_wrap(~Condition, scales = "free") +
  labs(
    title = "Histograms showing the distribution of Dopamine Values by Condition",
    x = "Values",
    y = "Count"
  )

fig.plot
```



2.4 Part d

Conduct the inferences they do in the paper. Make sure to report the results a little more comprehensively – that is your parenthetical should look something like: ($t = 23.99$, $p < 0.0001$; $g = 1.34$; 95% CI: 4.43, 4.60). Ensure to verify any conditions for use.
Note: Your numbers will vary slightly because they reported two-sided test p-values instead of the specified one-sided tests.

- "The close responses differed significantly from 0 ($p = 1.63 \times 10^{-8}$).''
  Note: Use the appropriate one-sided test.
- "The far responses differed significantly from 0 ($p = 5.17 \times 10^{-8}$).''
  Note: Use the appropriate one-sided test.
- "The difference between populations was significant ($p = 1.04 \times 10^{-8}$).''
  Note: Use the two-sided test.

Solution:

used left-tailed test for further, right-tailed test for closer, and two-sided test for difference in the t-tests. I then obtained the t-statistics, p-values, hedge-g effect sizes, and confidence intervals for each.

```r
library(effectsize)

# t-tests
closer <- t.test(x = fig.dat$Closer, mu = 0, alternative = "greater") # increases when singing closer

further <- t.test(x = fig.dat$Further, mu = 0, alternative = "less") # decreases when singing further

difference <- t.test(x = fig.dat$Difference, mu = 0, alternative = "two.sided") # differences btw closer and further

n <- nrow(fig.dat) # number of values for all

# t-statistics for each
(closer_t.stat <- mean(fig.dat$Closer) / (sd(fig.dat$Closer) / sqrt(n))) # for closer
```

```
[1] 9.157253
```

```r
(further_t.stat <- mean(fig.dat$Further) / (sd(fig.dat$Further) / sqrt(n))) # for further
```

```
[1] -7.924497
```

```r
(difference_t.stat <- mean(fig.dat$Difference) /
  (sd(fig.dat$Difference) / sqrt(n))) # for differences
```

```
[1] 9.14276
```

```r
# p-values
(p.closer <- pt(closer_t.stat, df = n - 1, lower.tail = FALSE)) # right tailed for closer
```

```
[1] 1.33263e-09
```

```r
(p.further <- pt(further_t.stat, df = n - 1)) # left tailed for further
```

```
[1] 1.865612e-08
```

```r
#two-sided for difference
p.low <- pt(difference_t.stat, df = n - 1)
p.high <- pt(difference_t.stat, df = n - 1, lower.tail = FALSE)
(p.difference <- 2 * min(p.low, p.high))
```

```
[1] 2.746058e-09
```

```r
# hedges-g (for effect sizes)
(g_closer <- hedges_g(x = fig.dat$Closer, mu = 0)) # for closer
```

```
Hedges' g |       95% CI
------------------------
1.77      | [1.14, 2.39]
```

```r
(g_further <- hedges_g(x = fig.dat$Further, mu = 0)) # for further
```

```
Hedges' g |         95% CI
--------------------------
-1.53     | [-2.10, -0.95]
```

```r
(g_difference <- hedges_g(x = fig.dat$Difference, mu = 0)) # for differences
```

```
Hedges' g |       95% CI
------------------------
1.77      | [1.14, 2.39]
```

Results

I verified that the observations are independent and that the distributions don't show very strong skewness or outliers, making the t-procedures appropriate.

($t = 9.157253$, $p < 0.0001(1.33263e - 09)$; $g = 1.77$; 95% CI: 1.14, 2.39).

The further responses were significantly less than 0 ($t = -7.924497$, $p < 0.0001(1.865612e - 08)$; $g = -1.53$; 95% CI: -2.10, -0.95).

The difference between conditions was significant ($t = 9.14276$, $p < 0.0001(2.746058e - 09)$; $g = 1.77$; 95% CI: 1.14, 2.39).

2.5 Part e

For each hypothesis test in the previous question, plot the null T distribution and the observed t statistic. Ensure to use `geom_ribbon()` to highlight the rejection region and the p-value. Additionally, add the resampling distribution for T to the chart to demonstrate any differences between the null distribution and the distribution suggested by the data.

Solution:

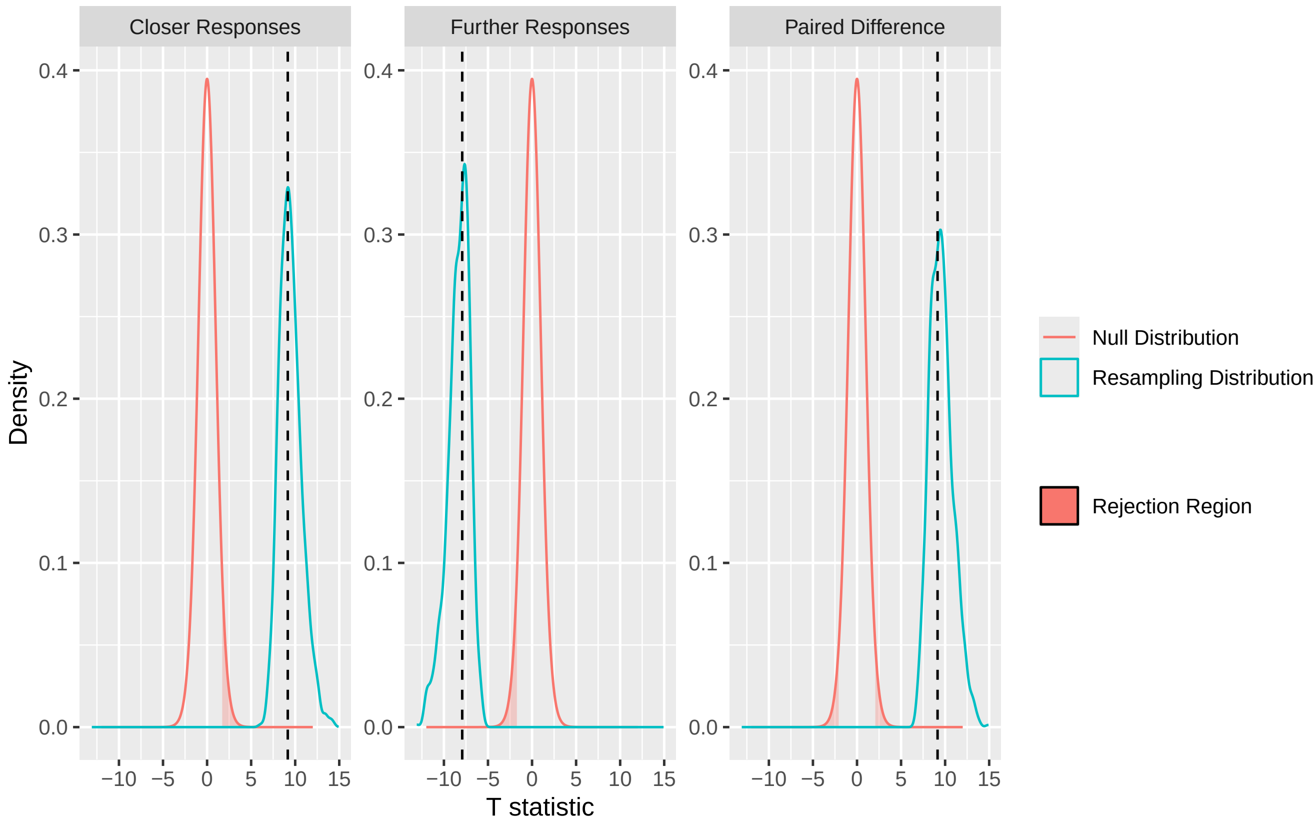
Used a for-loop to run the plotting process for the closer, further, and paired-difference tests by selecting each column, calculating observed t-statistic, and generating the null t-distribution. Used resampling to create a distribution of the t-statistic for each and stored the null distributions, resampling results, and observed statistics in separate tibbles. I then plotted the null distributions, shaded the rejection regions with `geom_ribbon()`, added the resampling distributions with `geom_density()`, and marked the observed t-statistics with dashed vertical lines.

```r
set.seed(123)

# defaults
alpha <- 0.05
mu0 <- 0

# variables with categories
tests <- c("Closer Responses", "Further Responses", "Paired Difference")
alts <- c("greater", "less", "two.sided")

# tibbles outside loop
null_all <- tibble()
resamp_all <- tibble()
tstat_all <- tibble()

for (i in 1:3) {
  # number for each section
  if (i == 1) {
    x <- fig.dat$Closer
```

```r
  } else if (i == 2) {
    x <- fig.dat$Further
  } else {
    x <- fig.dat$Difference
  }
  n <- length(x) # size of observations
  df <- n - 1 # degree of freedom
  xbar <- mean(x) #mean
  s <- sd(x) # std
  t.stat <- (xbar - mu0) / (s / sqrt(n)) #t-stat

  # t_critical values
  right <- qt(1 - alpha, df)
  left <- qt(alpha, df)
  lower <- qt(alpha / 2, df)
  upper <- qt(1 - alpha / 2, df)

  #x-vals for distribution
  t_vals <- seq(-12, 12, length.out = 1000)
  dens <- dt(t_vals, df) #density

  # tibble for all
  null_df <- tibble(
    test = tests[i],
    t = t_vals,
    density = dens
  )
  if (alts[i] == "greater") {
    null_df <- null_df |> mutate(reject = t >= right) # for right tailed
  } else if (alts[i] == "less") {
    null_df <- null_df |> mutate(reject = t <= left) # for left tailed test
  } else {
    null_df <- null_df |> mutate(reject = t <= lower | t >= upper) # for two.tailed test
  }

  # resampling
  ts <- rep(NA, 1000)
  for (j in 1:1000) {
    samp <- sample(x, n, replace = TRUE)
    ts[j] <- (mean(samp) - mu0) / (sd(samp) / sqrt(n))
  }

  # join to outside tibbles
  null_all <- bind_rows(null_all, null_df)
  resamp_all <- bind_rows(resamp_all, tibble(test = tests[i], t = ts))
  tstat_all <- bind_rows(tstat_all, tibble(test = tests[i], t_stat = t.stat))
}

## plotting
ggplot(null_all, aes(x = t, y = density)) +
  geom_line(aes(color = "Null Distribution")) + # for null
  geom_ribbon(
    # for one-tailed tests
    data = null_all |> filter(test != "Paired Difference", reject),
    aes(ymin = 0, ymax = density, fill = "Rejection Region"),
    alpha = 0.3
  ) +
  geom_ribbon(
    #left-side of two.tailed
    data = null_all |>
      filter(test == "Paired Difference", t <= qt(0.025, df = n - 1)),
    aes(ymin = 0, ymax = density, fill = "Rejection Region"),
    alpha = 0.3
  ) +
  geom_ribbon(
    # right side of two.tailed
    data = null_all |>
      filter(test == "Paired Difference", t >= qt(0.975, df = n - 1)),
    aes(ymin = 0, ymax = density, fill = "Rejection Region"),
    alpha = 0.3
  ) +
  geom_density(
    # resampling distrib
    data = resamp_all,
    aes(x = t, color = "Resampling Distribution"),
    inherit.aes = FALSE
  ) +
  geom_vline(
    # for t-statistic
    data = tstat_all,
    aes(xintercept = t_stat),
    linetype = "dashed"
  ) +
  facet_wrap(~test, scales = "free_y") +
```

```
  labs(
    #labels
    x = "T statistic",
    y = "Density",
    color = "",
    fill = ""
  )
```

Figure 1: Null t-distributions, observed t-statistics, rejection regions, and resampling distributions.



References

Champely, Stephane. 2020. *Pwr: Basic Functions for Power Analysis*. https://CRAN.R-project.org/package=pwr.

Kasdin, Jonathan, Alison Duffy, Nathan Nadler, Arnav Raha, Adrienne L Fairhall, Kimberly L Stachenfeld, and Vikram Gadagkar. 2025. "Natural Behaviour Is Learned Through Dopamine-Mediated Reinforcement." *Nature*, 1–8.